Light

GCSE Computer Science — Memory and Storage

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A computer has both RAM and a hard disk drive.
Why does a computer need secondary storage as well as RAM?

- ○ **A.** Secondary storage is faster than RAM for running programs
- ○ **B.** RAM loses its contents when power is turned off, so secondary storage is needed for permanent data
- ○ **C.** Secondary storage is used to perform calculations instead of the CPU
- ○ **D.** RAM can only store text files, while secondary storage can store images

Q2 (1 mark)

A traditional hard disk drive (HDD) stores data using magnetic technology.
How is data physically stored on a magnetic hard disk?

- ○ **A.** By burning tiny holes into the surface of the disk with a laser
- ○ **B.** By using coloured dyes that change between light and dark
- ○ **C.** By trapping electrons in microscopic cells that retain charge
- ○ **D.** By magnetising tiny areas on spinning platters to represent binary 1s and 0s

Q3 (1 mark)

CDs, DVDs, and Blu-ray discs all use optical storage technology.
How does an optical drive read data from these discs?

- ○ **A.** By detecting magnetic patterns on the disc surface
- ○ **B.** By measuring electrical resistance in different areas of the disc
- ○ **C.** By reading electronic chips embedded within the disc
- ○ **D.** By shining a laser beam and detecting reflections from pits and lands on the disc surface

Q4 (1 mark)

Solid state drives (SSDs) and USB flash drives use solid state storage technology.
How is data stored in solid state storage?

- ○ **A.** By magnetising areas on a spinning disk
- ○ **B.** By recording sound waves onto a crystal surface

- ○ **C.** By creating physical holes in a plastic surface
- ○ **D.** By trapping electrical charge in microscopic cells called flash memory cells

Q5 (1 mark)

Computer storage is measured using standard units.
How many bits are in one byte?

- ○ **A.** 2 bits
- ○ **B.** 4 bits
- ○ **C.** 16 bits
- ○ **D.** 8 bits

Q6 (1 mark)

A nibble is a unit of digital data.
How many bits are in one nibble?

- ○ **A.** 4 bits
- ○ **B.** 8 bits
- ○ **C.** 16 bits
- ○ **D.** 32 bits

Q7 (1 mark)

A student wants to email a large video file to a friend, but the file is too big to attach.

What technique could reduce the file size so it can be sent?

- ○ **A.** Encryption
- ○ **B.** Formatting
- ○ **C.** Defragmentation
- ○ **D.** Compression

Q8 (1 mark)

A music streaming service uses lossy compression for its audio files. What is a characteristic of lossy compression?

- ○ **A.** The original file can be perfectly reconstructed from the compressed version
- ○ **B.** Lossy compression only works on text files
- ○ **C.** The file size increases after compression
- ○ **D.** Some data is permanently removed, so the original cannot be fully restored

Q9 (1 mark)

A programmer compresses a software file before distributing it to users. Why would lossless compression be essential for software files?

- ○ **A.** Lossless compression makes the software run faster
- ○ **B.** Lossless compression is the only type that works on binary files
- ○ **C.** Lossless compression adds security features to protect the software

○ **D.** Lossless compression allows the exact original file to be restored, ensuring the software works correctly

Q10 (1 mark)

A laptop manufacturer chooses to use an SSD instead of an HDD.
What is one advantage of solid state storage over magnetic storage?

○ **A.** SSDs have larger storage capacity than HDDs

○ **B.** SSDs are cheaper per gigabyte than HDDs

○ **C.** SSDs have no moving parts, making them more durable and faster to access data

○ **D.** SSDs use magnetic fields to store data more reliably

Q11 (1 mark)

A business needs to store 20 TiB of archived data as cheaply as possible.
Which type of secondary storage would typically be most cost-effective for this purpose?

○ **A.** USB flash drives

○ **B.** Blu-ray discs

○ **C.** Magnetic hard disk drives (HDDs)

○ **D.** Solid state drives (SSDs)

Q12 (1 mark)

A software company distributes its product on DVD-ROM discs.

What advantage does optical storage offer for software distribution?

- ○ **A.** DVDs can store more data than any other storage medium
- ○ **B.** DVDs use magnetic technology that is more reliable than other methods
- ○ **C.** DVDs are faster than solid state storage for installing software
- ○ **D.** DVDs are cheap to manufacture, portable, and can be read on most computers

Q13 (2 marks)

A file has a size of 4 mebibytes (MiB).
How many kibibytes (KiB) is this equivalent to?

- ○ **A.** 4000 KiB, because 1 MiB = 1000 KiB
- ○ **B.** 4096 KiB, because 1 MiB = 1024 KiB
- ○ **C.** 32 KiB, because 4 MiB ÷ 128 = 32 KiB
- ○ **D.** 40 KiB, because 4 MiB × 10 = 40 KiB

Q14 (2 marks)

A USB flash drive has a capacity of 8 gibibytes (GiB).
How many mebibytes (MiB) can it store?

- ○ **A.** 8000 MiB
- ○ **B.** 8192 MiB
- ○ **C.** 64 MiB

○ **D.** 8.192 MiB

Q15 (2 marks)

A student is choosing between an SSD and an HDD for their new computer.
Which statement correctly compares these two storage types?

○ **A.** HDDs are faster than SSDs because spinning disks can move data more quickly than electronic circuits

○ **B.** SSDs are more durable than HDDs because SSDs have no moving parts that can be damaged by physical shock

○ **C.** HDDs are more expensive per gigabyte than SSDs because magnetic technology is newer

○ **D.** SSDs are less reliable than HDDs because electronic components fail more often than mechanical parts

Q16 (2 marks)

A photographer needs to compress their images.
When would lossy compression be acceptable, and when would lossless compression be essential?

○ **A.** Lossy is acceptable for sharing on social media; lossless is essential for professional editing where quality must be preserved

○ **B.** Lossy is essential for professional work; lossless is only for social media because file sizes do not matter online

○ **C.** Both types produce identical results, so it does not matter which is used

○ **D.** Lossless should always be used because lossy compression corrupts files beyond use

Q17 (2 marks)

Optical storage media come in different types: ROM, R, and RW.
Which description correctly explains the difference between CD-ROM, CD-R, and CD-RW?

○ **A.** CD-ROM can be written once; CD-R can be written multiple times; CD-RW is read-only

○ **B.** CD-ROM is read-only (manufactured with data); CD-R can be written once by the user; CD-RW can be erased and rewritten multiple times

○ **C.** All three types are identical and can be used interchangeably

○ **D.** CD-ROM holds more data than CD-R, which holds more than CD-RW

Q18 (2 marks)

A hard disk drive (HDD) stores data on spinning platters.
How does the rotational speed of the platters affect performance?

○ **A.** Faster rotation increases storage capacity by fitting more data on each platter

○ **B.** Faster rotation reduces data access time because the read/write head reaches data more quickly

○ **C.** Slower rotation improves reliability because there is less mechanical stress

○ **D.** Rotational speed has no effect on performance; only the number of platters matters

Q19 (2 marks)

A computer uses both RAM and secondary storage.
Which statement correctly describes the difference in volatility between these storage types?

○ **A.** Both RAM and secondary storage are volatile and lose data when power is removed

○ **B.** RAM is non-volatile and retains data; secondary storage is volatile and loses data

○ **C.** RAM is volatile and loses data without power; secondary storage is non-volatile and retains data

○ **D.** Volatility depends on the manufacturer, not the type of storage

Q20 (2 marks)

An image file is 2.5 MiB in size. A user has 10 MiB of free storage space. How many copies of this image can fit in the available space?

○ **A.** 2 copies, because 10 ÷ 2.5 = 4 but you cannot have partial files

○ **B.** 4 copies, because 10 ÷ 2.5 = 4 exactly

○ **C.** 5 copies, because 10 × 0.5 = 5

○ **D.** 25 copies, because 10 × 2.5 = 25

Q21 (2 marks)

A hospital stores patient medical records that must remain completely accurate.
Which compression method must be used for these records, and why?

- ○ **A.** Lossy compression, because it produces smaller files that are faster to transfer
- ○ **B.** No compression, because compression always introduces errors into data
- ○ **C.** Lossless compression, because it preserves all original data exactly when decompressed
- ○ **D.** Either type is acceptable because modern compression never loses important data

Q22 (2 marks)

Solid state drives have a limited number of write cycles before cells begin to fail.
How does this affect the suitability of SSDs for different applications?

- ○ **A.** SSDs should never be used because they will fail immediately
- ○ **B.** SSDs are ideal for applications with frequent writes because writing data extends their lifespan
- ○ **C.** SSDs are best for applications that mostly read data, with fewer write operations to preserve lifespan
- ○ **D.** Write cycle limits only apply to magnetic storage, not solid state

Q23 (2 marks)

Storage capacity is measured using a hierarchy of binary units.
Which sequence correctly orders these units from smallest to largest?

- ○ **A.** Byte, nibble, kibibyte, mebibyte, gibibyte
- ○ **B.** Bit, nibble, byte, kibibyte, mebibyte, gibibyte, tebibyte
- ○ **C.** Kibibyte, mebibyte, gibibyte, byte, bit
- ○ **D.** Bit, byte, nibble, kibibyte, mebibyte

Q24 (3 marks)

A bitmap image has the following properties:
• Resolution: 800 × 600 pixels
• Colour depth: 24 bits per pixel
What is the uncompressed file size of this image?

- ○ **A.** 1,440,000 bytes, because 800 × 600 × 3 = 1,440,000
- ○ **B.** 480,000 bytes, because 800 × 600 = 480,000 pixels
- ○ **C.** 11,520,000 bytes, because 800 × 600 × 24 = 11,520,000
- ○ **D.** 14,400,000 bytes, because 800 × 600 × 24 × 10 = 14,400,000

Q25 (3 marks)

A video streaming service stores 2 tebibytes (TiB) of content.
How many gibibytes (GiB) is this equivalent to?

- ○ **A.** 2000 GiB, because 1 TiB = 1000 GiB
- ○ **B.** 2048 GiB, because 1 TiB = 1024 GiB
- ○ **C.** 2,097,152 GiB, because 1 TiB = 1024 × 1024 GiB

- ○ **D.** 0.00195 GiB, because 2 ÷ 1024 = 0.00195

Q26 (3 marks)

A sound file has the following properties:
• Sample rate: 44,100 samples per second
• Bit depth: 16 bits per sample
• Duration: 3 minutes
• Mono (single channel)
What is the uncompressed file size in bytes?

- ○ **A.** 15,876,000 bytes, because 44,100 × 16 × 180 = 127,008,000 bits = 15,876,000 bytes
- ○ **B.** 2,116,800 bytes, because 44,100 × 16 × 3 = 2,116,800
- ○ **C.** 127,008,000 bytes, because 44,100 × 16 × 180 = 127,008,000
- ○ **D.** 7,938,000 bytes, because 44,100 × 180 = 7,938,000 samples

Q27 (3 marks)

An original file is 24 MiB. After compression, it is 6 MiB.
What is the compression ratio, and how much space is saved?

- ○ **A.** Compression ratio 4:1; space saved 18 MiB (75%)
- ○ **B.** Compression ratio 1:4; space saved 6 MiB (25%)
- ○ **C.** Compression ratio 6:24; space saved 30 MiB (125%)
- ○ **D.** Compression ratio 24:6; space saved 4 MiB (16.7%)

Q28 (3 marks)

A video production company needs storage for editing 4K video projects. Requirements include:

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• Fast read/write speeds for editing
• At least 2 TiB capacity
• Reliability for professional work
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Which storage solution best meets these requirements?

- ○ **A.** Multiple DVD-RW discs, because they can be rewritten and provide reliable backup
- ○ **B.** A large capacity HDD, because magnetic storage is the most reliable and fastest option
- ○ **C.** A high-capacity SSD, because solid state storage offers fast speeds and reliability with no moving parts
- ○ **D.** USB flash drives, because they are portable and can be swapped when full

Q29 (3 marks)

A backup drive has 500 GiB of free space. A user wants to back up files totalling 520,192 MiB.
Will all the files fit on the drive?

- ○ **A.** Yes, because MiB is smaller than GiB so 520,192 MiB must be less than 500 GiB
- ○ **B.** Yes, because 520,192 ÷ 1024 = 508 GiB, which rounds down to 500 GiB
- ○ **C.** No, because 520,192 MiB = 508 GiB, which exceeds the 500 GiB available
- ○ **D.** No, because MiB and GiB cannot be compared directly

Q30 (3 marks)

A school is considering different storage technologies for various purposes:

• Purpose A: Distributing educational software to students

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• Purpose B: Storing the school's database with frequent read/wri
• Purpose C: Archiving old records that are rarely accessed
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Which combination of storage technologies is most appropriate?

- ○ **A.** A: DVD-ROM; B: SSD; C: HDD
- ○ **B.** A: SSD; B: HDD; C: DVD-ROM
- ○ **C.** A: HDD; B: DVD-RW; C: SSD
- ○ **D.** A: USB flash drive; B: DVD-ROM; C: SSD

Total: **55** marks • Time: **30 min**

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